Building a Sample for Time Series Analysis

For the Inference of Macro Models

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Summer School on Macroeconomic Modelling and Policy

Objectives

- Downloading data in real time in MATLAB;
- Getting a stationary sample for macroeconomic analysis.

Additional reading list

- [Nelson and Plosser, 1982]
- [Canova, 2007]
- [Smets and Wouters, 2007]
- [Uhlig, 1995]

Course materials (slides, code, problem set):

`github.com/Vermandel/Macro-Modelling-Course-Nancy`

Back in time

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- Influential revolution, including in macroeconomic theory: pushed toward much more parsimonious models (Lucas' Revolution).
- Development of a rich set of time series models: AR, MA, VAR, ARCH, ... up to DSGE models (that we will estimate later on).

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- As new data are released, applied economists must re-run their forecasting models taking into account the latest observation released.
- In recent times, this policy work operates with real-time data to facilitate the repetitive work of the econometrician.
- We will use DB·nomics to get data in real time.

Outline

Core assumption in conventional models

- A time series Y_t is a process in sequence over time $t \in \{1, 2, ..T\}$:

$$\{y_1, y_2, ..., y_T\}$$

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- Understanding how this sample was generated is a fundamental task for an applied economist.
- By identifying the **underlying data-generating process**, we can develop meaningful quantitative analyses (e.g. causality analysis, forecasting, counterfactuals, etc).

Core assumption in conventional models

- One can assume a specific **Data Generating Process** (i.e. a time series model) with a specific distribution for prediction errors. These models typically relate the present value of a series to past values (AR model) and past prediction errors (MA):

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- ... but under certain conditions (stationarity, Gaussian errors, linearity) the computation of $E\left(y_{t+h}|I_t\right)$ is **straightforward**.
- In this class, our focus will be only on **linear Gaussian time series models**, which means we must be careful about the features of the sample we utilize.

Core assumption in conventional models

But which kind of features should we look for?

- **Stationarity** (weak definition) typically allows the mean function $E(y_t)$ and auto-covariance function $E[(y_s - E(y_s))(y_t - E(y_t))]$ to be time independent.

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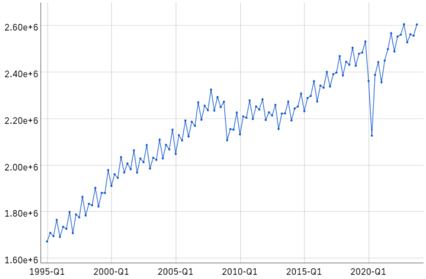
- One can estimate the parameters of the time series model $\hat{f}\left(\cdot\right)$, and provide out-of-sample forecasts assuming the distribution of y_t **remains unchanged** over the forecasting horizon (very convenient).
- Suppose our time series model is $y_t = \mu + \phi y_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim N(0, \sigma^2)$. One can estimate $\hat{\mu}$, $\hat{\phi}$ and $\hat{\sigma}$, and compute the mean-wise forecast $E\left(y_{t+1} | I_t\right) = \hat{\mu} + \hat{\phi} Y_t$ and quantify uncertainty: $E\left(y_{t+1} | I_t\right) \pm 1.96 \cdot \hat{\sigma}$.

But in practice...

Most of time series are affected by trend and seasonal components that do **not** imply (weak) stationarity. See GDP time series:



Gross domestic product at market prices – Euro area

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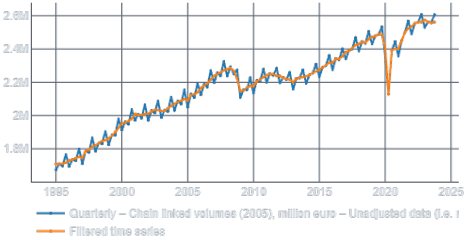
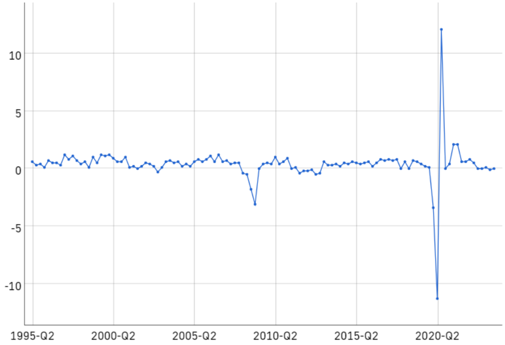
   - Use first difference filtering (e.g., express in growth rates)
   - Use a business cycle filter (HP, bandpass, etc)
   - Eventually divide by working age population to get rid off low frequency component

First deseasonalize:



Deseasonalized gross domestic product at market prices – Euro area

Then impose stationarity $\Delta \log Y_t$



Deseasonalized GDP growth at market prices – Euro area

DBnomics: Real-Time Data for Forecasting

What is DBnomics?

DBnomics is a platform that aggregates economic datasets from various public sources.

Why is it useful for forecasting?

- Provides access to a vast array of real-time economic data.
- Allows economists and analysts to track economic indicators as they are released.
- Enables more accurate and timely forecasting models by incorporating the latest data.

DBnomics: Real-Time Data for Forecasting

Scenario: INSEE French GDP Data Release

- INSEE releases French GDP data every first Monday of the month at 10:00 AM.
- As a forecaster, you only need to rerun your code at 10:01 AM to access the latest data.
- This allows you to quickly generate graphs and draft reports for management.

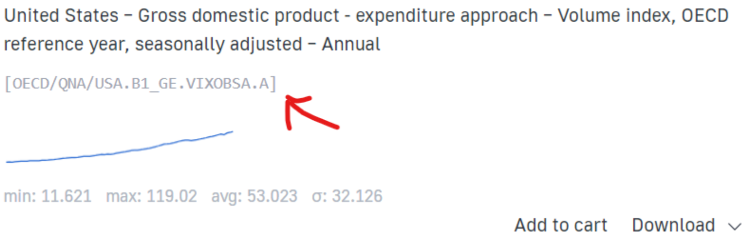
Using DBnomics

- Access the latest French GDP data from INSEE via the DBnomics API.
- Integrate the new data into your forecasting model.
- Generate updated graphs and reports for management with minimal delay.

DBnomics: Real-Time Data for Forecasting

How to use DBnomics?

1. Go to db.nomics.world
2. Browse among all the available databases, find your time series, and pick its token (a unique identifier for each series):



DBnomics: Real-Time Data for Forecasting

How to use DBnomics?

3 Download the DBnomics MATLAB plugin call_dbnomics.m

4 Query DBnomics in real time (assuming you have an internet connection):

```
[output_mat,output_table,dates_nb] =  call_dbnomics(varargin);
```

Input:

`varargin`: string tokens list ('token1','token2',...)

Outputs:

`output_mat`: $T \times (N+1)$ matrix of extracted data, 1st column is date vector in MATLAB format.

`output_table`: same matrix but in table format.

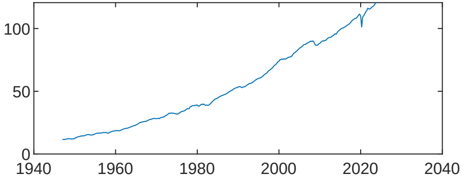
`dates_nb`: vector of dates in numeric values.

DBnomics: Real-Time Data for Forecasting

How to use DBnomics?

Example:

```
[output_mat,output_table,dates_nb] =  call_dbnomics('OECD/QNA/
    USA.B1_GE.VIXOBSA.Q');
figure;
plot(dates_nb,output_mat(:,2))
```



DBnomics: Real-Time Data for Forecasting

Building my sample for the US

Suppose a New Keynesian model comprising 3 core variables $[y_t, \pi_t, r_t]$. Our gross data from DBnomics comprise price data $\mathbb{P}_t$, GDP (volume) $\mathbb{Y}_t$ and nominal rate $\mathbb{R}_t$. One can simply impose for the two stationary variables the following transformation:

$$G_{\mathbb{Y}t} = \log(\mathbb{Y}_t) - \log(\mathbb{Y}_{t-1})$$
$$G_{\mathbb{P}t} = \log(\mathbb{P}_t) - \log(\mathbb{P}_{t-1})$$

One can next map the model and the data as follows:

$$\begin{aligned} y_t - y_{t-1} &= G_{\mathbb{Y}t} \\ \pi_t &= G_{\mathbb{P}t} \\ \underbrace{r_t}_{\text{model}} &= \underbrace{\mathbb{R}_t}_{\text{data}} \end{aligned}$$

DBnomics: Real-Time Data for Forecasting

Let's have a look at the code...

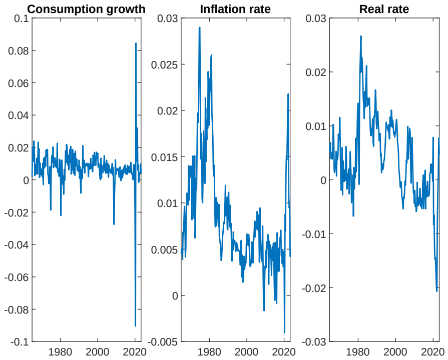
```
[raw_table,~,dates_nb] = call_dbnomics(...
  'OECD/QNA/USA.B1_GE.CQRSA.Q',...
  'OECD/QNA/USA.B1_GE.DNBSA.Q',...
  'OECD/KEI/IR3TIB01.USA.ST.Q');

valid_idx = all(~isnan(raw_table(:,2:end)),2);
P = raw_table(valid_idx,3);
P = P / P(find(floor(dates_nb(valid_idx)) == 2015,1));
gy_obs = diff(log(raw_table(valid_idx,2) ./ P));
pi_obs = diff(log(P));
r_obs  = raw_table(valid_idx,4) / 400;
```

Volume GDP identifies real activity, the deflator gives inflation, and the nominal rate is used directly in the sample.

DBnomics: Real-Time Data for Forecasting

Building my sample for the US

Run `my_db_US.m` to get the model observables:



A quick recap for your report:

- We will employ linearized and stationary macro models, which implies to have a sample distribution consistent with our model.
- Check first that your data do not exhibit seasonal component $\rightarrow$ use deseasonalized series or applies methods to remove seasonal component (e.g. X13 method).
- Check your sample is stationary (visual screening to assess the data have no upward trend as for GDP). If so, employ first order difference.
- If stationary and no seasonal features, your sample is ready for applied work.

A general formulation

- Consider a structural model written in a state-space form:

$$\mathbb{E}_t\{f_\Theta\left(y_{t-1}, y_t, y_{t+1}, \eta_t\right)\} = 0$$

where:

- y_t vector $N_y \times 1$ of endogenous variables;
- η_t vector $N_\eta \times 1$ of gaussian innovations, $\eta_t \sim \mathcal{N}(0, \Sigma)$.
- $f_\Theta()$ set of nonlinear equations.
- Θ vector of structural parameters.
- $\mathbb{E}_t\{\cdot\}$ the expectation scheme.

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- Spoiler alert: we will use the 3rd one!

A general formulation

- Consider a function $f(x)$. A Taylor expansion around a fixed point a reads as:

$$f(x) \simeq f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ldots$$

- Two natural questions:
  1. Which order of approximation to consider?
  2. Which value of a?

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- F, G, H and M are matrices stacking first order derivatives of f_Θ
- $\hat{z}_t = y_t - a$ is typically expressed in distance from steady state a=y. Why? a linear model combined with Gaussian stochastic shocks $\eta_t \sim \mathcal{N}(0, \Sigma)$ implies that $y_t - y \sim \mathcal{N}(0, \Sigma_y)$. The smallest deviations $E((y_t - a)^2)$ are obtained when 1st order expansion computed around steady state.

A first order Taylor expansion

- One looks for a recursive solution that would locally:

$$\hat{z}_t = P\hat{z}_{t-1} + Q\eta_t$$

  where P and Q are two unknown matrices.
- Appendix: derivation of P and Q

History of modern macroeconometrics

- Back in the 2 000s, dominance of time series models (e.g. VAR) to make forecasts and historical decomposition.
- DSGE models were not yet fully exploited on empirical grounds to compete with a-theoretical models.
- Estimation of DSGE models were based on weak information method: matching moments.
- Few attempts to use full-information techniques: [Fair and Taylor, 1983] & [Ireland, 2004], but subject to identification issues.

Principles of likelihood techniques

- Consider a specific Data Generating Process described by our DSGE model:

$$\hat{z}_t = P(\theta)\,\hat{z}_{t-1} + Q(\theta)\,\eta_t \text{ with } \eta_t \sim \mathcal{N}(0, \Sigma)$$
$$Y_t = H(\theta)\,\hat{z}_t + v_t, \text{ with } v_t \sim \mathcal{N}(0, R)$$

  - Where $\theta \subset \Theta$ is the subset of parameters to be estimated, the remaining being calibrated.
  - Subset of observable variables $Y_t \subset \hat{z}_t$, recognized as the set of macroeconomic time series, and map the model's equations into observables, where v_t are measurement errors.

Principles of likelihood techniques

- Consider a sample Y_1, Y_2 ,..., Y_T; one can compute a prediction error:

$$S_t(\theta) = Y_t - E_{t-1}\{Y_t(\theta)\}$$

where the prediction errors variance $\Omega_t = E(S_t S_t')$.

- Unlike VAR, in DSGE there are more endogenous variables than observable ones $\rightarrow$ use the Kalman filter to update recursively by picking the 'best' prediction through optimal mean squared error (MSE) estimator.

- The likelihood function is given by:

$$\log p(Y_{1:T}|\theta) = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{1}^{T}\log(|\Omega_t|) - \frac{1}{2}\sum_{1}^{T} S_t{}'\Omega_t^{-1}S_t \qquad (1)$$

Principles of likelihood techniques

- So in principle, (i) compute prediction errors (i.e. sequence of shocks) $S_{\theta,t}$ that replicates a sample $Y_{1:T}$, (ii) calculate likelihood function, (iii) relative comparison of $\log p(Y_{1:T}|\theta')$ against $\log p(Y_{1:T}|\theta'')$ allows us to determine which θ'/θ'' is the most likely to have generated the sample.
- Optimization can be implemented to determine θ_{MLE} that is the value of θ in the population of parameters that is the most likely to have generated the sample:

$$\min_{\{\theta\}} -\log p(Y_{1:T}|\theta)$$

- As long as $\log p(Y_{1:T}|\theta)$ is differentiable in θ, one can use gradient based methods to optimize.

Bayesian techniques

Limits of likelihood techniques

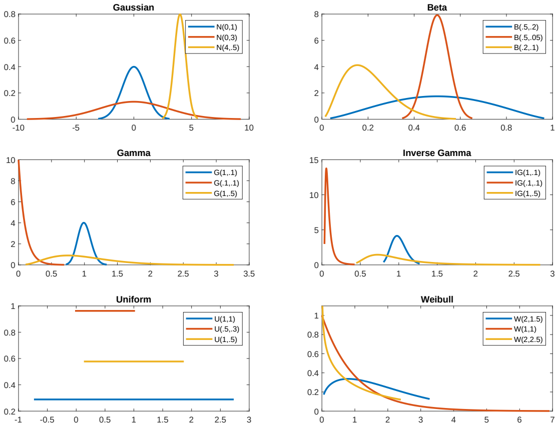
- Any model is an approximation of the true data-generating process.
- Does it matter? If the data-generating distribution does not belong to the model's set of probability distributions, then the model is misspecified.
- **Some illustrations:** our current medium-scale NK model does not include a financial sector or epidemic dynamics $\rightarrow$ recessions will be captured erroneously by wrong sources of fluctuations.
- Likelihood function will be higher for implausible parameter values to accommodate the wrong representation of the data given by our misspecified model.

Setting Priors

- Dynare includes a set of priors, each prior is summarized by a shape (e.g. gaussian, beta, etc.) and a mean and standard deviation.
- How to adjust my prior? Usually have a look at already published papers, or use common sense.

Shape	Name	Support	Example(s)
normal_pdf	$\mathcal{N}(\mu, \sigma)$	$(-\infty, +\infty)$	Policy parameters, utility curvatures.
gamma_pdf	$\mathcal{G}(\mu, \sigma)$	$(0, +\infty)$	Elasticities, utility curvatures, trends.
beta_pdf	$\mathcal{B}(\mu, \sigma)$	$[0, 1]$	Probabilities, AR-MA terms, shares.
inv_gamma_pdf	$\mathcal{IG}_1(\mu, \sigma)$	$(0, +\infty)$	Shocks standard deviations.
uniform_pdf	$\mathcal{U}(\mu, \sigma)$	$(-\infty, +\infty)$	Set lower/upper bounds.
weibull_pdf	$\mathcal{W}(\mu, \sigma)$	$[0, +\infty)$	Measurement error std.
inv_gamma2_pdf	$\mathcal{IG}_2(\mu, \sigma)$	$(0, +\infty)$	Standard deviations.

Setting Priors



Estimation steps:

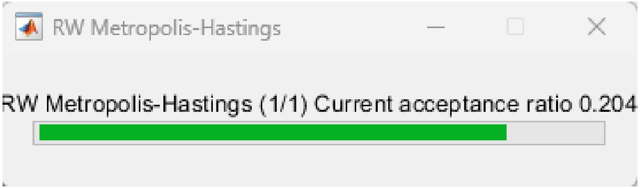
1. Maximize $p(\theta|Y_{1:T})$ using numerical optimization techniques to find the mode of the distribution.
2. Use numerical integration to compute confidence intervals (parametric uncertainty) through Metropolis Hastings algorithm.

Each iteration requires: for a given candidate θ to compute prediction errors $\{S_{\theta,t}\}_1^T$, evaluate sample likelihood + posterior.

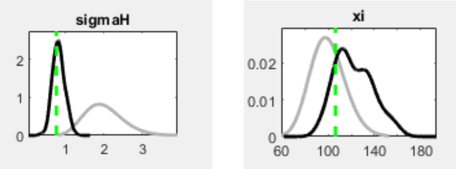
Dynare automates those steps, making life easier!

Acceptance

- Metropolis-Hastings explores the parameter space. Sampling: $\theta_i^{cand} \sim \mathcal{N}(\theta_{i-1}^{cand}, c \times \Sigma_\theta)$ with parameter c that scales the size of the random perturbation around the mode.
- c must be set to match 20-30% of the acceptance rate.
- If the acceptance rate is too low: the chain is stuck in a local minimum. If too high: chain does not explore parameter space sufficiently.



Parameter Identification



- Dark line: posterior distribution $p(Y_{1:T}|\theta)p(\theta)$, while gray line is $p(\theta)$ for alternative parameter values θ. Green line is mean.
- Left figure: data are informative (data dominate); right figure: data are uninformative (prior dominates).
- Identification implies that prior does not dominate posterior. Here, we should calibrate ξ and re-run estimation.

Toy DSGE: model and data

Economic structure

We use the simplest possible two-equation log-linearised system:

$$\underbrace{gc\_obs_t}_{\text{cons. growth}} = \sigma \underbrace{rr\_obs_t}_{\text{real rate}}, \qquad rr\_obs_t = \rho_r \, rr\_obs_{t-1} + \varepsilon^r_t.$$

σ is the slope of the IS curve; ρ_r governs persistence of the real rate.

- Observable 1: **consumption growth** (`gc_obs`), built from real consumption in `my_db_US.m`.
- Observable 2: the **real rate** (`rr_obs`), proxied by $r^r_t \approx r_t - \pi_{t+1}$.
- Estimated parameters: ρ_r, σ, σ_{ε^r}, and a measurement-error std on `gc_obs`.

Toy DSGE: priors and estimation

Estimated parameters

In `simple_euler_real_rate.mod`, four parameters are estimated: ρ_r, σ, σ_{ε^r}, and the measurement-error std on `gc_obs`.

- ρ_r: **Beta**($\mu = 0.80$, $\sigma = 0.10$) — persistence in $[0, 1]$, real rate is usually persistent.
- σ: **Gamma**($\mu = 2$, $\sigma = 0.20$) — IS slope, strictly positive, tight prior around 2.
- σ_{ε^r}: **inv-Gamma**($\mu = 0.01$, $s = 2$) — structural shock std, strictly positive.
- $\sigma_{\texttt{gc\_obs}}$: **Weibull**($\mu = 0.001$, $\sigma = 0.001$) — measurement-error std on consumption growth.

Pedagogical objective: connect data treatment, observables, priors, and likelihood-based estimation in the smallest possible Dynare example.

A simple Dynare exercise

```
var gc_obs rr_obs;
varexo eps_r;
parameters rho_r sigma;

rho_r = 0.8;  sigma = 1.0;

model(linear);
  gc_obs = sigma * rr_obs;
  rr_obs = rho_r * rr_obs(-1) + eps_r;
end;

varobs gc_obs rr_obs;
```

```
estimated_params;
  rho_r,       beta_pdf, 0.8,   0.10;
  sigma,       gamma_pdf, 2,    0.20;
  stderr eps_r,  inv_gamma_pdf, 0.01,  2;
  stderr gc_obs, weibull_pdf, 0.001, 0.001;
end;

estimation(datafile=myobs,
  prefilter=1, mode_compute=4,
  mh_replic=0, diffuse_filter);
```

Takeaways

Data

- Check for **stationarity** (visual screening + ADF test); apply first-difference or HP filter if needed.
- Check for **seasonality**; use a seasonal filter (e.g. X-13ARIMA-SEATS) if present.
- Use **real-time data** (e.g. DBnomics) to update the dataset in real time.

Estimation

- Set **priors** consistent with the economic literature and calibration targets.
- Monitor the **acceptance rate** of the Metropolis-Hastings sampler (target: 20–30%).
- Check **parameter identification**: posterior should dominate the prior.

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Rational expectations: deriving P and Q

Setup

Substitute $\hat{z}_t = P\hat{z}_{t-1} + Q\eta_t$ into $F\mathbb{E}_t\{\hat{z}_{t+1}\} + G\hat{z}_t + H\hat{z}_{t-1} + M\eta_t = 0$. Since $\mathbb{E}_t\{\hat{z}_{t+1}\} = P\hat{z}_t = P^2\hat{z}_{t-1} + PQ\eta_t$, matching coefficients gives two equations.

Quadratic equation for P

$$FP^2 + GP + H = 0$$

Rewrite as a $2n \times 2n$ generalised eigenvalue problem with companion pencil

$$\underbrace{\begin{bmatrix} F & 0 \\ 0 & I_n \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} -G & -H \\ I_n & 0 \end{bmatrix}}_{B}$$

Bayesian techniques

Bayes' Theorem

$$\underbrace{p(\theta|Y)}_{\text{posterior}} = \frac{\overbrace{p(Y|\theta)}^{\text{likelihood}}\ \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(Y)}_{\text{marginal density}}}$$

- $p(Y|\theta)$: probability of observing the data given parameters — the **likelihood function**, evaluated via the Kalman filter.
- $p(\theta)$: **prior** distribution on θ, encoding beliefs before seeing the data.
- $p(\theta|Y)$: **posterior** distribution — what we learn about θ after observing Y.
- $p(Y) = \int p(Y|\theta)\, p(\theta)\, d\theta$: **marginal data density**, a normalising constant

Out-of-Sample Forecasts

- Governments and central banks rely on DSGE model forecasts to design monetary and fiscal policies
- A forecast provides a baseline scenario, and allows us to introduce some policies to evaluate how the forecast is affected

A quick recap. Suppose that we have picked $\hat{\theta}$ with highest posterior

$$\hat{z}_t = P\left(\hat{\theta}\right)\hat{z}_{t-1} + Q\left(\hat{\theta}\right)\eta_t$$

with corresponding estimated sequence of shocks $\{\eta_t\}_{t=1}^{T}$ for given $\hat{\theta}$, one wants to make some out-of-sample forecasts over horizon $T + h$

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